
Intelligence Engineering

Solving Real World Problems At Scale

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LECTURE _

Orientation

what this course is, what it assumes, what you are going to know after.

You have been given a lot of raw **data**,

**and you want to engineer the shit
out of it.**

You have been given a business problem,

**and you want to choose the right
metrics and objectives to solve it.**

Your initial models perform well in offline experiments,

and you want to **deploy them at scale.**

You want to harden your deployments

**to quickly detect, debug, and
address issues that arise in
production.**

There is a lot of manual work and handling involved so far,

**and you want to operation vacation
your machine learning product or
service.**

You want to learn about a framework and foundation for building machine learning systems

**that can be shared and reused
across machine learning use cases
and data products.**

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Overview

Machine Learning and Data Engineering

The lecture in one slide

Lecture	Contents
I	Overview of Machine Learning Systems
II	Introduction to Machine Learning Systems Design
III	Data Engineering Fundamentals
IV	Training Data
V	Feature Engineering
VI	Model Development and Offline Evaluation
VII	Model Deployment and Prediction Service
VIII	Data Distribution Shifts and Monitoring
IX	Continual Learning and Test in Production
X	Infrastructure and Tooling for MLOps
XI	The Human Side of Machine Learning

The lecture in one slide

I Overview of Machine Learning Systems

The small part and the big picture
Understanding Machine Learning Systems

II Introduction to Machine Learning Systems Design

Business and ML Objectives
Requirements for ML Systems
Iterative Process
Framing ML Problems
Mind Versus Data

III Data Engineering Fundamentals

Data Sources
Data Formats
Data Models
Data Storage Engines and Processing
Modes of Dataflow
Batch Processing Versus Stream Processing

The lecture in one slide

IV Training Data

Sampling

Labeling

Class Imbalance

Data Augmentation

V Feature Engineering

Learned Features Versus Engineered Features

Common Feature Engineering Operations

Data Leakage

Engineering Good Features

VI Model Development and Offline Evaluation

Model Development and Training

Model Offline Evaluation

The lecture in one slide

VII Model Deployment and Prediction Service

Machine Learning Deployment
Myths
Batch Prediction Versus Online Prediction
Model Compression
ML on the Cloud and on the Edge

VIII Data Distribution Shifts and Monitoring

Causes of ML System Failures
Data Distribution Shifts
Monitoring and Observability

IX Continual Learning and Test in Production

Continual Learning
Test in Production

The lecture in one slide

X Infrastructure and Tooling for MLOps

Storage and Compute
Development Environment
Resource Management
ML Platform
Build Versus Buy

XI The Human Side of Machine Learning

User Experience
Team Structure
Responsible AI

Course Work

Presentation (15 min) + Q&A (open)

You get recordings of a bee hive entrance, and you are to contribute by data engineering, an annotation policy, a prediction service or specialized tooling.

Open whiteboard, three slides or one page only. Be ready when the class starts.

The bee



Course Work

What I am looking for ..

Introduction by Motivation and Literature

Method by Explanation and or Demonstration

Results show what was achieved and how well

Discussion on findings, limitations and what's next

Smart and Aesthetic Thoughts, Designs, Ideas and **Questions**.

The bee

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Extra Mile

Special projects

If you are in for an extra mile, let me know. I have special projects that I can assign to you.

Wrapped in the same presentation and Q&A format, yes it will be about Bananas, and yes it will be more challenging, more time and effort.

Classic loose loose short term, yet **win long term**.

A large banana





By the end of this lecture,

**Get yourself a presentation slot
and brainstorm a project idea to
work on.**

References

- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.